

SATQUERY AI — SIH 26167

Competitive Gap, Research Landscape & Differentiation Blueprint

A practical research document for building a solution that goes beyond the mandatory SIH requirements

Item	Details
Problem ID	26167
Problem	SatQuery AI – An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries
Organization	Indian Space Research Organisation (ISRO)
Department	Department of Space / ISRO
Category	Software
Theme	Space Technology
Purpose of this document	Identify what already exists, locate the real implementation/research gaps, and define a differentiated feature set for the SIH solution.

Executive Summary

The central mistake to avoid in SIH 26167 is treating the project as only a satellite-image chatbot. The official problem already requires a natural-language interface, remote-sensing adaptation, VQA, an additional single-image capability, bi-temporal change analysis, optical–SAR analysis, and agentic model/tool orchestration. Therefore, these capabilities should be treated as the baseline rather than the main novelty.

Current commercial and research systems show that natural-language geospatial interaction, automated change detection, and agentic geospatial workflows are already emerging. Google Earth AI supports natural-language queries over Earth observation imagery; Planet provides automated building/road detection and change detection and has introduced agentic geospatial AI; askEarth provides a geospatial intelligence agent; and recent research explores tool-augmented remote-sensing assistants. Therefore, the team should differentiate through the reliability and decision layer around those capabilities rather than claiming that the basic agent itself is new.

Recommended direction: build SatQuery AI as an Evidence-Verified Adaptive Remote-Sensing Agent. The agent should not only choose a model and return an answer; it should assess input suitability, understand ambiguity, choose an analysis path, cross-check multimodal/temporal evidence, detect disagreement, estimate uncertainty, and abstain or request better evidence when the result is not trustworthy.

1. What the SIH Problem Actually Requires

Requirement	Meaning	Priority
Remote-sensing adaptation	At least one visual/VLM	Mandatory

	component must be fine-tuned or otherwise adapted using BigEarthNet.txt or open-source training data.	
Single-image VQA	Answer natural-language questions about one optical/multispectral or SAR image.	Mandatory
Additional single-image task	Captioning/scene description OR text-guided region grounding.	Mandatory
Bi-temporal analysis	Change description or change-based VQA for two corresponding images at different times.	Mandatory
Optical–SAR analysis	Joint information extraction from co-registered optical/multispectral and SAR imagery.	Mandatory
Agentic orchestration	Interpret query, validate inputs, select tools/models, execute workflow, combine outputs and expose an execution trace.	Mandatory
Evidence and confidence	Return visual evidence, confidence information and auditable execution information.	Expected

2. The Key Strategic Insight: Baseline vs Differentiation

For SIH, a feature can be technically impressive but still weak as a novelty claim if the problem statement already explicitly asks for it. The team should divide the product into two layers.

Layer	Examples	How to present it
SIH baseline	VQA, captioning/grounding, change analysis, optical–SAR, agentic routing, GeoTIFF support	Required implementation
Differentiation layer	Input reliability assessment, ambiguity handling, evidence verification, cross-model disagreement, uncertainty/abstention, evidence chain, adaptive re-analysis	Research/innovation contribution

3. Competitive Landscape — What Already Exists

3.1 Google Earth AI

Google's Earth AI material describes specialized AI models for finding objects, classifying scenes and monitoring changes across Earth observation imagery, with natural-language querying through Gemini

Enterprise Agent Platform. This means natural-language Earth observation search and agentic access to specialized imagery models are already part of the commercial landscape.

Implication for your team: do not claim 'natural-language satellite analysis' as the sole USP.

Source: Google AI — Earth AI.

3.2 Planet

Planet provides automated analytic feeds for building detection, road detection, vessel/aircraft detection and road/building change detection. Its change workflow uses segmentation and temporal aggregation and publishes structured outputs such as GeoJSON and GeoTIFF. Planet also announced an agentic geospatial AI experience that lets users ask questions and combines satellite data, AI change detection and public information.

Implication: object detection, infrastructure change detection, structured geospatial outputs and an agentic map/chat experience are not sufficient by themselves as a differentiator.

3.3 askEarth GIA

askEarth's GIA is presented as a geospatial intelligence agent that connects LLMs with geospatial data, algorithms, APIs and organizational data. It emphasizes natural conversation, interactive maps and domain-aware answers.

Implication: 'an LLM that selects geospatial tools' is also not a strong standalone novelty.

3.4 Research Direction: Tool-Augmented Remote-Sensing Agents

Recent remote-sensing research is moving toward vision-language agents that can schedule or orchestrate specialist tools for tasks such as VQA, captioning, detection, grounding and segmentation. This reinforces the conclusion that the basic multi-tool agent architecture is becoming a research direction rather than a unique invention.

Implication: your innovation should move one layer deeper—from tool selection to trustworthy tool selection, evidence verification, uncertainty handling and safe decision-making.

4. Competitive Gap Analysis

Capability	Landscape	Novelty potential	Recommended use
Natural-language query	Already available	Do not use as main USP	Use as interface
Specialist model routing	Already emerging	Weak as sole novelty	Make routing data-aware and auditable
Change detection	Mature commercial capability	Not unique	Add change reliability and false-change analysis
Object/region detection	Widely available	Not unique	Use as evidence for higher-level reasoning
Interactive map/chat	Already available	Not unique	Use for usability
Optical + SAR support	Technically active research area	Potentially differentiating in the exact SIH workflow	Make fusion plus disagreement handling central
Bi-temporal analysis	Widely researched/commercialized	Not unique	Add alignment and temporal reliability

			checks
Confidence score	Common	Weak if just model probability	Build a multi-factor reliability score
Evidence-grounded answer	Increasingly common	Useful but not enough alone	Connect evidence to data quality and model agreement
Abstention / insufficient evidence	Important research direction	Strong practical differentiator	Make it a first-class system outcome
Cross-model disagreement handling	Less commonly exposed to end users	Strong candidate	Use as a visible verification layer
Input-quality-aware planning	Underdeveloped compared with basic routing	Strong candidate	Make it the first gate before model execution

5. Recommended New Concept

Evidence-Verified Adaptive Analysis

The proposed concept is to turn SatQuery AI from an answer-generating agent into an evidence-aware analysis system. The agent first determines whether the available data can support the requested analysis, then selects the appropriate specialist workflow, and finally verifies whether the produced result is consistent with spatial, temporal and multimodal evidence.

Core pipeline

1. Natural-language query understanding
2. Task and input requirement inference
3. Data Reliability Gate
4. Specialist model/tool selection
5. Modality-specific preprocessing
6. Primary inference
7. Evidence extraction
8. Cross-check / verification
9. Reliability and uncertainty assessment
10. Answer, warning, re-analysis or abstention
11. Auditable evidence chain

6. Feature 1 — Data Reliability Gate

Before any expensive AI inference, the system should determine whether the input is scientifically suitable for the requested operation. This is especially important because remote-sensing images are not ordinary JPEGs: they may contain multiple bands, projections, spatial transforms, acquisition metadata and large dimensions.

Check	Example	Failure consequence
Format	GeoTIFF/TIFF	Unsupported processing
Band information	Optical/multispectral bands	Wrong model or interpretation
CRS	Coordinate reference system	Geographic mismatch
Resolution	Different pixel sizes	False spatial comparison
Spatial extent	Same area	Invalid pair analysis

Registration	Pixel/feature alignment	False change
Acquisition metadata	Date/sensor	Wrong temporal reasoning
Image quality	Cloud/noise/invalid pixels	Unreliable inference

7. Feature 2 — Data Reliability Score

Instead of returning only 'valid/invalid', calculate a structured reliability assessment. For example, the system can combine metadata completeness, alignment quality, resolution compatibility and image-quality indicators into a pre-analysis score.

Illustrative output:

- CRS compatibility: PASS
- Spatial overlap: PASS
- Registration quality: 91%
- Metadata completeness: 94%
- Image quality: 82%
- Overall input reliability: 88% — suitable for requested analysis

Important: the exact scoring formula should be defined and experimentally validated; it should not be presented as a scientifically meaningful probability unless calibrated.

8. Feature 3 — Query Ambiguity Resolver

Natural-language queries can be incomplete. Instead of forcing the model to guess, SatQuery should identify missing information and either ask a clarification question or state a safe assumption.

User query	Problem	System response
Find buildings	No target area/classification intent	Ask whether all buildings or newly constructed buildings are required
What changed?	No explicit change type	Infer available temporal tasks and state the comparison scope
Find water	No spatial/temporal scope	Ask whether the user wants all water or changed water extent
Has the city grown?	Requires a definition of growth	Interpret as built-up expansion and disclose the assumption

9. Feature 4 — Evidence Verification Loop

The primary model's output should not automatically become the final answer. A verification layer should check whether the answer is supported by the available visual/spatial evidence.

Stage	Question
Prediction	What did the specialist model detect?
Localization	Where exactly is the evidence?
Temporal check	Does the evidence correspond between dates?

Modality check	Do optical and SAR signals support the interpretation?
Quality check	Could alignment, cloud, noise or resolution explain the result?
Agreement check	Do independent specialist outputs agree?
Decision	Verified, uncertain, re-analyze or abstain?

10. Feature 5 — Cross-Modal Disagreement Detection

Optical and SAR should not always be forced into a single confident answer. When the two modalities disagree strongly, the system should expose the disagreement and investigate possible causes.

Example:

- Optical interpretation: water, confidence 84%
- SAR interpretation: non-water, confidence 73%
- System detects a modality disagreement
- System checks registration, resolution and acquisition metadata
- If uncertainty remains high, the final answer is marked uncertain rather than forced into a binary label

11. Feature 6 — Real Change vs Apparent Change

A temporal difference is not automatically a real-world change. Apparent differences can arise from registration errors, seasonal effects, atmospheric conditions, shadows, sensor differences, resolution differences or SAR-specific noise.

The proposed Change Reliability Analyzer should classify the result as, for example: likely genuine change, probable change but uncertain, or insufficient evidence.

Outcome	Meaning
Verified change	Evidence is consistent and quality checks pass
Likely change	Evidence is strong but one or more uncertainty factors remain
Possible false change	Detected difference may be explained by acquisition/alignment effects
Insufficient evidence	The system cannot responsibly support the claim

12. Feature 7 — Cross-Model Verification

For high-impact queries, the agent can use one specialist model for the primary prediction and another model for verification. For example, a change detector may identify a candidate region and a grounding/segmentation model can verify whether the region actually corresponds to the claimed object or land-cover type.

This should be selective rather than mandatory for every query, because chaining models increases latency and resource use.

13. Feature 8 — Evidence Chain

Every answer should be traceable through a compact evidence chain without exposing private internal reasoning.

- User query
- Input images and detected modalities
- Task selected
- Preprocessing performed
- Models/tools executed
- Primary prediction
- Spatial/temporal evidence
- Verification result
- Reliability/confidence information
- Final answer and limitations

14. Feature 9 — Safe Abstention

The system must have three legitimate outcomes: answer confidently, answer with an explicit warning, or refuse to make a claim because the evidence is insufficient.

State	Example
CONFIDENT	Two well-aligned images; strong change evidence; verification passes
UNCERTAIN	Change detected but seasonal or registration uncertainty remains
INSUFFICIENT EVIDENCE	Only one image is provided for a question requiring temporal comparison

15. Feature 10 — Adaptive Compute / Evidence Acquisition

Satellite images can be extremely large. Rather than applying every model to the full image, the system can use a coarse-to-fine strategy: broad screening, candidate region selection, high-resolution analysis only where relevant, and final evidence verification.

This can reduce GPU memory use and latency while making the agent's processing path more explainable.

16. Proposed Differentiated Architecture

12. 1. User Query Layer — receives the natural-language request and image inputs.
13. 2. Query Understanding Layer — converts the request into a structured task specification.
14. 3. Data Reliability Gate — validates format, modality, metadata, spatial/temporal compatibility and quality.
15. 4. Agent Planner — chooses a workflow from the model/tool registry.
16. 5. Preprocessing Layer — performs modality-aware normalization, tiling, registration or reprojection when justified.

17. 6. Specialist Models — VQA, captioning/grounding, change analysis, optical–SAR analysis.
18. 7. Evidence Layer — extracts masks, boxes, coordinates, temporal comparisons and model scores.
19. 8. Verification Layer — checks evidence consistency, disagreement and quality risks.
20. 9. Reliability Engine — combines validated signals into a calibrated reliability/uncertainty assessment.
21. 10. Response Composer — generates the natural-language answer and limitations.
22. 11. UI/Audit Layer — displays imagery, evidence, reliability, selected tools and execution trace.
23. 12. Monitoring Layer — records latency, failures and model versions while minimizing sensitive data exposure.

17. Example: A Strong End-to-End Scenario

User asks: “Has the built-up area increased between these two images, and where?”

24. Agent identifies a bi-temporal change-analysis task.
25. Input gate checks that the images correspond spatially and contain usable metadata.
26. Registration/quality checks produce a reliability assessment.
27. Change model identifies candidate changed regions.
28. A grounding/segmentation component localizes the candidate regions.
29. The verification layer checks whether the regions are consistent with built-up expansion rather than obvious alignment or acquisition artifacts.
30. The system calculates a calibrated reliability/uncertainty result.
31. The UI displays before/after images, change mask, affected region, answer, reliability and execution trace.
32. If evidence is weak, the system explicitly reports uncertainty instead of producing an unsupported claim.

18. What NOT to Claim as Your USP

- “We use an LLM to analyze satellite images.”
- “We have a chatbot for satellite images.”
- “We use multiple AI models.”
- “We perform change detection.”
- “We support optical and SAR.”
- “We have an agent.”
- “We show confidence.”
- “We have a map interface.”

These are useful capabilities, but several are explicitly required by the SIH statement or are already present in commercial/research systems. They should be implemented as the baseline and combined into the stronger reliability-oriented contribution.

19. What to Claim Instead

Recommended positioning: “SatQuery AI is an evidence-verified adaptive remote-sensing agent. Beyond routing specialist models from natural-language queries, it evaluates whether the supplied imagery is suitable for the requested analysis, cross-checks temporal and cross-modal evidence, detects model disagreement, estimates uncertainty, and can abstain when the evidence does not support a reliable conclusion.”

20. Comparison: Basic SatQuery vs Differentiated SatQuery

Dimension	Basic implementation	Recommended implementation
Query	Natural-language query	Natural-language query + ambiguity detection
Routing	Select one model	Select workflow based on task + input quality
Input	Upload and validate format	Scientific/geospatial reliability gate
Change	Detect difference	Separate genuine change from possible acquisition/alignment effects
Optical-SAR	Fuse modalities	Fuse + expose and resolve modality disagreement
Confidence	Model probability	Calibrated multi-factor reliability where feasible
Answer	Text result	Answer + evidence + limitations
Failure	Error message	Warning, re-analysis, clarification or abstention
Audit	Basic logs	Structured evidence chain and execution trace
Compute	Run full image	Coarse-to-fine / tile-aware adaptive processing

21. Evaluation Plan for the Differentiation

Metric area	Measure	Purpose
Task accuracy	VQA/caption/grounding/change metrics	Measure core capability
Routing accuracy	Correct vs incorrect tool selection	Test positive and negative queries
Input quality detection	Correctly accept/reject/flag pairs	No silent geospatial errors
Alignment robustness	Performance under controlled shifts	Avoid false changes
Cross-modal disagreement	Correctly flag conflicting evidence	Avoid forced wrong answers
Abstention quality	Correct abstention on unsupported cases	Reduce unsafe claims
Evidence localization	IoU/box/mask quality where applicable	Make answers verifiable
Domain shift	Benchmark vs target-like data	Measure generalization
Latency	End-to-end time	Keep interactive workflows usable
Resource use	RAM/VRAM/concurrency	Prevent crashes

22. SIH Demo Strategy

A normal demo should show a successful answer. A stronger SIH demo should show both success and intelligent failure handling.

Demo	What to show	Why it matters
Normal VQA	Correct answer with evidence	Mandatory baseline

Bi-temporal change	Before/after/change map	Mandatory baseline
Optical-SAR	Joint analysis	Mandatory baseline
Ambiguous query	System asks/declares assumption	Shows safe language interaction
Poor pair	System detects alignment/quality issue	Shows scientific reliability
Model disagreement	System exposes conflict and lowers reliability	Shows trustworthy multimodal reasoning
Insufficient evidence	System abstains instead of hallucinating	Strong government-use-case behavior

23. Implementation Priority

Priority	Component	Implementation goal
P0	Mandatory SIH baseline	VQA + one additional single-image task + change + optical-SAR + agent routing
P0	Input reliability gate	CRS, dimensions, bands, metadata, pair compatibility
P0	Evidence output	Masks/boxes/regions/model IDs where available
P1	Query ambiguity	Clarification and safe assumptions
P1	Change reliability	Alignment/quality-aware change interpretation
P1	Cross-model disagreement	Optical vs SAR / primary vs verifier
P1	Abstention	Insufficient evidence outcome
P2	Adaptive compute	Coarse-to-fine region selection and caching
P2	Advanced reporting	Downloadable evidence/audit report

24. Risks and How the New Design Addresses Them

Risk	Mitigation
Wrong model selected	Routing tests + explicit tool contracts + negative examples
False temporal change	Registration/quality checks + change verification
Optical-SAR mismatch	Modality-aware preprocessing + disagreement detector
Hallucinated answer	Evidence requirement + abstention
Large GeoTIFF crash	Tiling + bounded memory + adaptive compute
Benchmark overfitting	Separate benchmark and target-domain evaluation
Model incompatibility	Isolated model services and pinned dependencies
UI complexity	Progressive disclosure: simple answer first, evidence details on demand

25. Final Recommended Project Identity

Project category: Agentic multimodal remote-sensing intelligence system.

Core SIH capability: natural-language orchestration of specialist remote-sensing models.

Differentiation: evidence-verified adaptive analysis.

Hero feature: a reliability/evidence engine that decides whether a model result is sufficiently supported before presenting it as a conclusion.

Key user benefit: the user does not need to know which remote-sensing model to use, and the system does not pretend to know when the supplied evidence is inadequate.

26. Judge Questions You Should Be Ready For

Judge question	Recommended answer
“Google/Planet already do natural-language satellite analysis. What is different?”	Our differentiation is not natural-language access alone. We add an evidence-verified adaptive layer that evaluates input suitability, temporal/cross-modal consistency, model disagreement and uncertainty before accepting a conclusion.
“Why not just use one VLM?”	The SIH statement requires remote-sensing adaptation and multiple specialist capabilities. Different tasks such as grounding, temporal change and optical–SAR analysis require different specialized processing.
“What if your model gives the wrong answer?”	The system does not treat every model output as final. It requires task-appropriate evidence, checks quality and agreement, reports uncertainty, and can abstain when evidence is insufficient.
“How is your confidence score calculated?”	We will distinguish model confidence from system reliability and calibrate the final score using measurable signals such as input quality, evidence quality and model agreement. We will not claim a probability without calibration.
“What is your real research contribution?”	The contribution is the integration of query-driven specialist orchestration with data-quality-aware planning, evidence verification, disagreement handling and uncertainty-aware abstention for multimodal and bi-temporal remote-sensing workflows.
“Why is this useful for ISRO?”	Remote-sensing outputs can appear plausible even when spatial alignment, modality compatibility or acquisition conditions are poor. The proposed reliability layer is intended to reduce scientifically misleading results and make outputs auditable.

27. Research Boundaries — What You Should Not Overpromise

- Do not claim the entire architecture is globally novel; agentic geospatial systems already exist.
- Do not claim that evidence verification is completely absent from industry.
- Do not claim a TrustScore is scientifically valid until it is calibrated and experimentally evaluated.
- Do not promise perfect optical–SAR fusion; modality differences are intrinsically difficult.
- Do not promise that a public benchmark automatically predicts performance on the hidden ISRO/SAC dataset.
- Do not build many extra features before the mandatory SIH workflows are reliable.

28. Source-Based Conclusion

The supplied SIH problem and the team's existing risk analysis make clear that the hard part is an end-to-end pipeline: query → validation → preprocessing → model selection → inference → result validation → evidence → UI → evaluation. The existing risk document specifically identifies geospatial misalignment, optical–SAR incompatibility, temporal registration, hallucinated answers, domain shift, large-image performance and routing errors as high-priority risks. Therefore, the strongest direction is not to add another superficial model, but to make the orchestration layer aware of data quality and evidence reliability.

29. External Research References

- Google AI — Earth AI: natural-language Earth observation queries and specialized imagery models.
- Planet Documentation — Analytic Feeds and Road & Building Change Detection.
- Planet — Agentic AI for Earth Observation / geospatial intelligence.
- askEarth — GIA Geospatial Intelligence Agent.
- Recent remote-sensing agent/tool-augmentation literature, used here to establish that basic tool orchestration is an active research direction.

Prepared as a SIH 26167 competitive-gap and implementation research blueprint